Light

GCSE Computer Science - Cyber Security Prevention (Set 3)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is gait recognition as a biometric security measure?

- ○ **A.** Identifying people by their unique walking pattern and movement style
- ○ **B.** Scanning the sole of someone's foot
- ○ **C.** Measuring how fast someone can run
- ○ **D.** Checking if someone is wearing the correct shoes

Q2 (1 mark)

Why should passwords avoid using common words found in dictionaries?

- ○ **A.** Because dictionaries are too heavy to carry
- ○ **B.** Because common words are harder to remember
- ○ **C.** Because attackers use dictionary attacks that systematically try common words and phrases first
- ○ **D.** Because dictionaries contain spelling mistakes

Q3 (1 mark)

What is a 'honeypot' field in web forms, and how does it help prevent automated submissions?

- ○ **A.** A field that attracts bees to the website
- ○ **B.** A visible field asking users to type 'honey'
- ○ **C.** A hidden field that humans cannot see but bots automatically fill in, revealing themselves as automated
- ○ **D.** A sweet reward for completing the form correctly

Q4 (1 mark)

Why do email systems check if incoming emails actually came from the domain they claim to be from?

- ○ **A.** To detect spoofed emails where attackers fake the sender address to appear legitimate
- ○ **B.** To count how many emails are sent each day
- ○ **C.** To make emails arrive faster

- ○ **D.** To add colours to the email interface

Q5 (1 mark)

Why is it important that software updates also update the software's dependencies (other programs it relies on)?

- ○ **A.** To make the software more expensive
- ○ **B.** To make the software run slower
- ○ **C.** To use more storage space
- ○ **D.** Because vulnerabilities in dependencies can be exploited to attack the main software, even if the main software itself is secure

Q6 (1 mark)

Why do authenticator app codes change every 30 seconds?

- ○ **A.** To make the app more exciting to use
- ○ **B.** Because the phone battery runs out every 30 seconds
- ○ **C.** To ensure that intercepted or stolen codes become useless within seconds, limiting the window for attackers to use them
- ○ **D.** To test if users can type quickly enough

Q7 (1 mark)

What is sandboxing in computer security?

- ○ **A.** Filling a computer with sand to protect it
- ○ **B.** Running suspicious programs in an isolated environment where they cannot harm the rest of the system
- ○ **C.** A children's game played on computers
- ○ **D.** Removing all programs from a computer

Q8 (1 mark)

What is the difference between a firewall whitelist and a blacklist approach?

- ○ **A.** Whitelists block everything except approved items; blacklists allow everything except blocked items
- ○ **B.** Whitelists use white paper; blacklists use black paper
- ○ **C.** Whitelists are newer than blacklists
- ○ **D.** There is no difference between them

Q9 (1 mark)

Why do secure systems add a random 'salt' to passwords before storing them?

- ○ **A.** To prevent attackers from using pre-computed tables to crack multiple passwords at once, since each password has a unique salt
- ○ **B.** To make the password taste better
- ○ **C.** To make the password longer on screen

- ○ **D.** To add flavour to the database

Q10 (1 mark)

How does retina scanning differ from iris scanning for biometric authentication?

- ○ **A.** They are exactly the same technology
- ○ **B.** Retina scanning is done by looking at someone's elbow
- ○ **C.** Retina scanning uses sound waves; iris scanning uses light
- ○ **D.** Retina scanning analyses blood vessel patterns at the back of the eye, while iris scanning analyses the coloured ring around the pupil

Q11 (1 mark)

Why might anti-malware software offer a 'boot-time scan' that runs before Windows fully loads?

- ○ **A.** To scan the computer's shoes before it starts running
- ○ **B.** To make the computer boot faster
- ○ **C.** To detect and remove malware that hides itself once the operating system is running, but is visible during the boot process
- ○ **D.** To check if the computer is wearing its seatbelt

Q12 (1 mark)

What is 'tailgating' in physical security?

- ○ **A.** An unauthorised person following an authorised person through a secure door before it closes
- ○ **B.** Following too closely behind another car
- ○ **C.** Wagging your tail while waiting in a queue
- ○ **D.** Standing at the back of a queue

Q13 (2 marks)

Some websites use simple logic puzzles like 'What is 3 + 4?' instead of image-based CAPTCHAs. What are the advantages and limitations of this approach?

- ○ **A.** Logic puzzles are always better than images
- ○ **B.** They are accessible to screen readers and load faster, but simple maths is easily solved by bots, requiring more complex or varied questions to remain effective
- ○ **C.** Logic puzzles cannot be solved by humans
- ○ **D.** This approach has no limitations whatsoever

Q14 (2 marks)

What is continuous biometric authentication, and why might it provide better security than one-time login authentication?

- ○ **A.** It means typing your password continuously without stopping
- ○ **B.** It plays continuous music while you are logged in
- ○ **C.** It requires scanning your fingerprint every second

○ **D.** It continuously verifies the user throughout a session using behavioural patterns like typing rhythm and mouse movements, detecting if the original authenticated user is replaced by an imposter after login

Q15 (2 marks)

Why do some password managers warn you if your password appears in known data breaches?

○ **A.** Because if your password has been exposed in a breach of any site, attackers will try that same password on other sites - this practice of credential stuffing makes reused passwords dangerous even if your accounts were not directly breached

○ **B.** To sell you a new password

○ **C.** To make you feel bad about your password choices

○ **D.** Because all passwords eventually appear in breaches

Q16 (2 marks)

What security risk is created when software reaches 'end of life' and the vendor stops releasing updates?

○ **A.** The software becomes physically heavier

○ **B.** The software changes colour

○ **C.** Any new vulnerabilities discovered will never be patched, leaving users permanently exposed to attacks that exploit these flaws - the software becomes increasingly unsafe over time

○ **D.** End of life makes software run faster

Q17 (2 marks)

What is 'double opt-in' for email mailing lists, and what security benefit does it provide?

- ○ **A.** Clicking the subscribe button twice in a row
- ○ **B.** Having two different email addresses
- ○ **C.** Paying twice for a subscription
- ○ **D.** After signing up, users must click a confirmation link in their email before being added to the list - this prevents attackers from subscribing others to unwanted emails using fake sign-ups

Q18 (2 marks)

Why can having multiple recovery options for two-factor authentication actually reduce security?

- ○ **A.** Because recovery options always fail
- ○ **B.** Each recovery method is a potential bypass - if an attacker can compromise any single recovery option such as security questions, backup email, or phone number, they can disable 2FA entirely, making the overall system only as strong as its weakest recovery method
- ○ **C.** Recovery options make accounts impossible to access
- ○ **D.** There is no relationship between recovery options and security

Q19 (2 marks)

How does behavioural monitoring in modern anti-malware software differ from traditional file scanning?

- ○ **A.** File scanning checks files against known malware signatures before execution, while behavioural monitoring watches what programs actually do during execution - catching malware that evades signature detection by identifying malicious actions
- ○ **B.** Behavioural monitoring is louder than file scanning
- ○ **C.** Behavioural monitoring only works on Mondays
- ○ **D.** There is no difference - they work identically

Q20 (2 marks)

What is 'vishing' and how does it differ from phishing?

- ○ **A.** Vishing and phishing are the same thing
- ○ **B.** Vishing is voice-based phishing conducted over phone calls, where attackers impersonate trusted entities to extract sensitive information verbally - unlike email phishing which relies on fake websites and written messages
- ○ **C.** Vishing involves fishing for actual fish
- ○ **D.** Vishing only targets people whose names start with V

Q21 (2 marks)

What is 'liveness detection' in biometric systems and why is it important?

- ○ **A.** It checks if the biometric scanner is still working
- ○ **B.** It only allows users who are alive to use computers
- ○ **C.** It measures how lively and energetic the user appears

○ **D.** It determines whether the biometric sample comes from a living person present at that moment rather than a photograph, video recording, or artificial replica - preventing spoofing attacks that use fake biometric samples

Q22 (2 marks)

What is a DMZ (demilitarised zone) in network security?

○ **A.** A place where military personnel take breaks

○ **B.** A zone where computers do not work

○ **C.** A separate network segment between the public internet and internal network where public-facing servers are placed, limiting damage if those servers are compromised since attackers cannot directly reach internal systems

○ **D.** The area directly under a computer monitor

Q23 (2 marks)

Why do software vendors classify some updates as 'critical' and recommend immediate installation?

○ **A.** Because they want to interrupt users more often

○ **B.** Critical updates fix severe vulnerabilities that are actively being exploited or could allow complete system compromise - delaying these updates leaves systems exposed to known attacks with available exploits

○ **C.** Critical just means the update is larger in file size

○ **D.** All updates are equally important regardless of classification

Q24 (3 marks)

A dental practice protects patient records with these measures: (1) Staff log in with username, password, and fingerprint scan; (2) The system logs which staff member accessed which patient record and when; (3) Staff can only view records of patients they are treating that day. Analyse the security purpose of each measure.

- ○ **A.** Multi-factor authentication prevents unauthorised account access; audit logging creates accountability and enables investigation of inappropriate access; role-based access limits each person to only the data they need for their job - together these satisfy healthcare data protection requirements while protecting patient privacy
- ○ **B.** These measures only slow down staff without providing security
- ○ **C.** Only the fingerprint scan provides meaningful protection
- ○ **D.** Dental practices do not need security for patient records

Q25 (3 marks)

A small charity discovers issues with their computer security: (1) Volunteers share one 'volunteer' account because setting up individual accounts seems complicated; (2) The same password has been used for three years and most volunteers know it; (3) No one has administrator responsibility for updating software. Analyse how these practices create security vulnerabilities.

- ○ **A.** These are sensible practices for a small organisation with limited resources
- ○ **B.** Charities are never targeted by cyber attacks
- ○ **C.** Only the lack of updates creates a problem
- ○ **D.** Shared accounts eliminate individual accountability making incident investigation impossible; long-unchanged known passwords are vulnerable to former volunteers, spread through social circles, and likely appear in breach databases; no update responsibility means

vulnerabilities remain unpatched indefinitely - together creating severe unmanaged risk

Q26 (3 marks)

A homeowner installs smart home devices: smart locks, cameras, and thermostats. They all use the home WiFi network along with family computers and phones. Analyse the security implications of having IoT devices on the same network as personal devices.

- ○ **A.** Smart home devices are always completely secure
- ○ **B.** Keeping devices on separate networks would make them stop working
- ○ **C.** IoT devices often have weaker security than computers, receive fewer updates, and may have known vulnerabilities - if compromised, they provide attackers a foothold on the home network from which to attack other devices, potentially accessing personal data on computers and phones
- ○ **D.** The number of devices on a network has no security implications

Q27 (3 marks)

An email system implements multiple layers of protection for attachments: (1) Blocks executable file types like .exe and .bat; (2) Scans all attachments with anti-malware before delivery; (3) Opens attachments in a sandbox for analysis before allowing download. Analyse how each layer contributes to security and why multiple layers are used.

- ○ **A.** Only one layer of protection is needed
- ○ **B.** File type blocking stops obvious threats but is bypassed by disguised files; anti-malware scanning catches known threats but misses new malware; sandbox analysis detects malicious behaviour but

adds delay - multiple layers ensure that threats bypassing one defence are caught by another, providing defence in depth

○ **C.** These protections make email unusable

○ **D.** All three methods work identically

Q28 (3 marks)

An estate agent discovers that a laptop containing client data was stolen from a car. The laptop had: (1) Full disk encryption enabled; (2) A strong login password; (3) Remote wipe capability that was activated immediately after discovering the theft. Analyse how these protections limit the damage from the physical theft.

○ **A.** Physical theft of a device always means complete data compromise

○ **B.** These protections would have prevented the theft

○ **C.** Only remote wipe provides any protection

○ **D.** Full disk encryption means data is unreadable without the encryption key even if the hard drive is removed; the strong password prevents casual login attempts; remote wipe ensures that even if attackers eventually break encryption, the data will be erased first - together these provide layered protection turning a device theft into a hardware loss rather than a data breach

Q29 (3 marks)

Some organisations are moving toward 'passwordless authentication' using methods like biometrics, hardware keys, or magic links sent to verified devices. Analyse the potential advantages and challenges of eliminating traditional passwords.

○ **A.** Passwords are perfect and should never be replaced

○ **B.** Passwordless authentication has only disadvantages

○ **C.** Advantages include eliminating password-related attacks like phishing and credential stuffing, removing the burden of remembering complex passwords, and preventing password reuse; challenges include dependency on devices that can be lost or stolen, accessibility concerns, recovery complexity, and requiring users to adapt to unfamiliar methods

○ **D.** Removing passwords makes systems completely secure with no remaining vulnerabilities

Q30 (3 marks)

A security audit of a small business produces these findings: (1) Antivirus software is installed but virus definitions are 18 months out of date; (2) The WiFi network uses WEP encryption from 2005; (3) Employees have not received any security awareness training. If budget allows addressing only one issue immediately, analyse which should be prioritised and why.

○ **A.** All issues are equally urgent and none should be prioritised

○ **B.** The antivirus should be prioritised because software always matters most

○ **C.** The training should be prioritised because it is cheapest

○ **D.** The WEP encryption should be prioritised because it is a completely broken standard that can be cracked in minutes using freely available tools - it provides essentially no protection and allows anyone nearby to intercept all network traffic, while the other issues create risk but do not offer immediate trivial access

Total: **55** marks • Time: **30 min**

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